

How these sheets work

One sheet per competency, 6 sheets for this video. Each gives you two prompts beside one open box. You build one prompt in the box. The other is there if you want more practice before the exam.

Before you start

1. Pick two pens you can tell apart. Pass one is the course notes, before the video. Pass two is a second color, after the concept video, into the same box.
2. Stop pass one while part of the box is still empty. That empty part is what the video fills in.
3. Read your prompt first, then divide the box to fit it. Some prompts want one labeled drawing, some a grid, some three panels. The division is yours.
4. Words go inside the drawing, as labels and inside little boxes with arrows. A sentence running the width of the box does not count.

Why two prompts

A and B are never the same question twice. One usually asks for a labeled drawing and the other for a grid or a sequence, so whichever way you think, there is a way in. Neither is the easy option.

About the paper

You do not have to print this. Rule one box per competency onto your own paper, or set it up on a tablet, and write the prompt letter you chose at the top. Hand ruled counts exactly the same.

This is anatomy

Every prompt asks what a structure is, where it sits, and what it is next to. Function belongs to physiology and is not on these sheets.

The box is **drawn**, not written. Pick one prompt and divide the box however it needs. Words go inside the drawing, in labels and little boxes with arrows between them. Sentences running across the page do not count.

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COMP

WEEK 6 · ARTICULATIONS AND JOINTS

Structural and functional joint classification

You should be able to: Classify a joint by structure as fibrous, cartilaginous, or synovial and by function as synarthrosis, amphiarthrosis, or diarthrosis and state the criteria for each.

PICK ONE AND DO THAT ONE

☐ **A TWO-AXIS GRID**

Draw a grid with the three structural classes down the side and the three functional classes across the top. In every cell that exists in the body, draw one example joint and name it. Leave the impossible cells empty and cross them out.

Then: Write beside the grid the one thing the structural classes are sorted by, and the one thing the functional classes are sorted by.

☐ **B TWO SORTING TREES**

Draw two trees. Tree one starts with the question, what holds the bones together, and branches to fibrous, cartilaginous, synovial, each ending in a small drawing of that joint. Tree two starts with the question, how much does it move, and branches to synarthrosis, amphiarthrosis, diarthrosis, each ending in a body example.

Then: Draw a line between the two trees for every branch on one tree that always lands on the same branch of the other.

PASS 1 COLOR _____

PASS 2 COLOR _____

DRAW IT

split it up however the prompt needs

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Dr. Sharilyn Rennie

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WEEK 6 · ARTICULATIONS AND JOINTS

COMP

Fibrous and cartilaginous joints

You should be able to: Describe the three fibrous joint types and the two cartilaginous joint types by structure, mobility, and example.

PICK ONE AND DO THAT ONE

☐ **A FIVE LABELED DRAWINGS**

Draw the five joints as simple cross sections: suture, syndesmosis, gomphosis, synchondrosis, symphysis. In each, draw what sits between the two bones and label it, dense fibrous tissue, a ligament, a periodontal ligament, hyaline cartilage, or fibrocartilage. Write the body example under each.

Then: Mark each drawing with how much it moves, and circle the one that is a tooth in its socket.

☐ **B SORTING TABLE**

Draw a table with the five joint types down the side, grouped into fibrous and cartilaginous. Across the top: what connects the bones, how much it moves, its functional class, and one example. Draw the connecting tissue in the first column.

Then: Put a box around the two rows where the connecting tissue is cartilage and write which kind of cartilage each one uses.

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WEEK 6 · ARTICULATIONS AND JOINTS

COMP

Synovial joint structure and accessory parts

You should be able to: Identify the structures of a synovial joint, its accessory structures such as menisci, labra, bursae, and tendon sheaths, and its nerve and blood supply.

PICK ONE AND DO THAT ONE

☐ **A LABELED CROSS SECTION**

Draw a synovial joint cut through the middle. Label the articular cartilage on each bone, the joint cavity, the synovial membrane, the fibrous layer of the capsule, and the synovial fluid. Add a bursa outside the capsule, a tendon sheath around a tendon, and a meniscus inside the cavity, and label all three.

Then: Draw a nerve and a small artery reaching the capsule, and write which layer each one supplies.

☐ **B INSIDE-OUT HUB**

Put the joint cavity in the middle of the box. Spoke outward in layers: the fluid, the cartilage, the synovial membrane, the fibrous capsule, the ligaments. Then add a second ring of spokes for the accessory structures, meniscus, labrum, bursa, tendon sheath, each with a tiny drawing of where it sits.

Then: Write beside each accessory structure one joint in the body where you would find it.

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WEEK 6 · ARTICULATIONS AND JOINTS

COMP

Movements at synovial joints

You should be able to: Demonstrate and name the gliding, angular, rotational, and special movements permitted at synovial joints.

PICK ONE AND DO THAT ONE

☐ **A STICK FIGURE MOVEMENT SHEET**

Draw a stick figure for every movement: flexion and extension, abduction and adduction, circumduction, rotation, pronation and supination, dorsiflexion and plantar flexion, inversion and eversion, protraction and retraction, elevation and depression, opposition. Show the starting position and the moved position with an arrow between them, and label each.

Then: Group the figures into gliding, angular, rotational, and special, and draw a box around each group.

☐ **B PAIRED-OPPOSITES TABLE**

Draw a table with the movement pairs down the side and across the top: the joint where it happens, the plane it moves in, and a tiny stick-figure sketch. Fill the sketch column with drawings, and put the two members of each pair in one row so the opposites sit together.

Then: Circle the movements that have no opposite and write which joint each belongs to.

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WEEK 6 · ARTICULATIONS AND JOINTS

COMP

Six synovial joint types

You should be able to: Name the six synovial joint types with their axes of movement and give a body example of each.

PICK ONE AND DO THAT ONE

☐ **A SIX LABELED JOINT SHAPES**

Draw the six joint types as the shapes of their bone ends: plane, hinge, pivot, condylar, saddle, ball-and-socket. Draw the arrows for the way each one moves, and write the body example under each. Draw them in order from the least movement to the most.

Then: Mark each shape with how many axes it moves around, and draw the axes as lines through the joint.

☐ **B AXES SORT**

Draw three columns headed one axis, two axes, three axes. Put each of the six joint types in the right column with a small drawing of its shape and a body example. Add a fourth column for the one joint type that only glides.

Then: Beside each example joint, draw a tiny stick figure of the movement it allows.

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COMP

WEEK 6 · ARTICULATIONS AND JOINTS

Shoulder and knee joint structure

You should be able to: Describe the articulating bones and supporting ligaments of the glenohumeral and tibiofemoral joints and identify the rotator cuff, menisci, and cruciate and collateral ligaments.

PICK ONE AND DO THAT ONE

- ☐ **A TWO LABELED JOINTS**
Draw the shoulder from the front: the glenoid cavity, the head of the humerus, the labrum, the capsule, and the four rotator cuff tendons wrapping the head. Then draw the knee from the front with the patella removed: the femoral condyles on the tibial plateau, both menisci, the two cruciate ligaments crossing, and the two collateral ligaments on the sides.

Then: On each joint, circle the structure that holds the joint together most, and write what the joint depends on when that structure is torn.
- ☐ **B TWO-JOINT COMPARISON TABLE**
Draw a table with shoulder and knee across the top. Rows: the bones that meet, the shape of the joint, what deepens the socket, the ligaments that support it, the cartilage pads inside it, the muscles or tendons that stabilize it, and the direction it most often gives way. Draw the shape row instead of describing it.

Then: Draw a small side view of the knee under the table and label the cruciate ligaments so their crossing is visible.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

DRAW IT

split it up however the prompt needs